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Blockchain recordation and validation of video data

Abstract

A method of committing video data and associated metadata to a blockchain ledger includes preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model. The method also includes accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data, formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference, and accessing the preprocessed video data and associated metadata by a blockchain subscriber independent from the first artificial intelligence model. In response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, the method includes committing a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data.

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Background/Summary

PRIORITY CLAIM (1) The present application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/294,554 filed Dec. 29, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) This disclosure relates generally to validating the authenticity of video data and, more specifically, to the use of a blockchain/distributed ledger to record and validate the video data to prevent modification and/or manipulation of the data.

(2) Cameras are beneficial for use in many areas of commercial and personal practice. For example, security cameras are used within (and outside) commercial warehouses and on private personal property. Other applications use cameras along assembly lines for quality control purposes. With the increased capabilities of cameras having higher quality imagery (i.e., resolution) and a wider field of view, more area can be shown in the streaming video by the camera. A large portion of the frame/field of view may be of little or no interest to the consumer (e.g., a security or manufacturing company). However, current practices relay the entirety of the streaming video (i.e., the entire frame/field of view) to the consumer, which can be time and resource consuming due to the need to transfer large frame (i.e., field of view), high resolution video data.

SUMMARY

(3) A system and method for selection/extraction, preprocessing, and publishing of video data of a region of interest (i.e., a scene) that is a subset of a field of view of streaming video is disclosed herein. The system and method can also include processing the video data by a consumer/subscriber after the video data has been published.

(4) Streaming video data is received from a camera with a first field of view. The video data is then preprocessed, by a computer processor such as a gateway or digital container, according to preprocessing parameters defined within a runtime configuration file that is pushed down to the computer processor. The runtime configuration file can be stored and/or edited distant from the computer processor, and any edits/revisions to the runtime configuration file can be pushed to and applied by the computer processor to the streaming video data in real time to alter the preprocessing applied to the video data. The preprocessing can include formatting/cropping the streaming video data received from the camera so as to be first video data of a first region of interest (i.e., a scene) having a second field of view that is less than (shows less area than) the first field of view shown by the entirety of the streaming video data from the camera. The preprocessing as defined by the preprocessing parameters in the runtime configuration file can also include altering the first video data's grayscale, contrast, brightness, color threshold, size, blur, hue saturation value (HSV), sharpen, erosion, dilation, Laplacian image processing, Sobel image processing, pyramid up, and pyramid down (among others). The first video data can then be published to an endpoint (such as a topic on an asynchronous messaging library like ZeroMQ) for subscription and use by a first subscriber/consumer.

(5) The first video data can then be viewed, used, and/or processed by the first subscriber. The preprocessing as defined in the runtime configuration file can be tailored to the subscriber and the needs/uses of the subscriber and the processing to be performed by the subscriber. For example, the processing performed by the subscriber after publishing of the first video data may be using an artificial intelligence (AI) model to analyze scenarios occurring on/in the first video data. The AI model may require the first video data to be in a particular size, format, etc., which can be selected and applied during the preprocessing as set out in the runtime configuration file so that the

subscriber does not need to perform this preprocessing before applying the AI model. The processing, by a computer processor, of the first video data by the subscriber can be performed distant from the camera, the location at which the runtime configuration file is stored and/or edited, and the gateway/container upon which the preprocessing is performed. The first subscriber can perform the processing of the video data to determine at least one output, with the output being indicative of an inference dependent on the first video data. For example, the first video data can be processed by an AI model to determine the amount of a particular product that has passed by on an assembly line (i.e., the amount of the product being an inference dependent on the first video data). The processing can include other operations, such as applying optical character recognition, clipping the first video data to make a video having a specific duration, and/or capturing one frame from the first video data to create a static image of a specific moment of the first video data.

(6) The AI model can determine an output that is indicative of an inference, such as an output that is indicative that a particular product has passed through the field of view shown in the first video data. Whether the inference is correct may come under dispute. For example, a purchaser may dispute whether the manufacturer constructed the correct number of products (as counted by the AI model). In another example, a purchaser may dispute whether the product being constructed was without defects (as determined by the AI model). In such a situation, there may be a need to record the unaltered video data as well as the video data reviewed and potentially annotated by the AI model to provide unalterable, authenticated proof the inferences discovered/determined by the AI model. Recording the video data, annotated video data, and respective associated metadata can be performed by committing the data to a blockchain/distributed ledger, which creates an unalterable entry that can be referenced if the inferences or other information in the video data is disputed. Further entries can be committed to the blockchain ledger, such as an entry that includes video data for each inference that is discovered/determined by the AI model.

(7) One embodiment of a method of committing video data and associated metadata to a blockchain ledger includes preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model. The method also includes accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data, formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference, and accessing the preprocessed video data and associated metadata by a blockchain subscriber independent from the first artificial intelligence model. In response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, the method includes committing a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data.

(8) Another embodiment includes a system for authenticating video data and metadata associated with the video data. The system includes a blockchain ledger and a gateway, which includes a first computer processor, that is configured to preprocess the video data according to preprocessing parameters defined within a runtime configuration file to edit the video data to be suitable for use by a first artificial intelligence model. The first artificial intelligence model is configured to access the preprocessed video data and associated metadata, determine a first output that is indicative of a first inference dependent upon the preprocessed video data, and formulate first annotated video data and associated metadata that includes the first output indicative of the first inference. The system also includes a blockchain subscriber, which is executed by a second computer processor, that is independent from the first artificial intelligence model and is configured to access the preprocessed video data and associated metadata and a sink in communication with the first

artificial intelligence model and the blockchain subscriber. The sink is configured to, in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, commit a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic of the scene selection and preprocessing system.

(2) FIG. **2** is a schematic of the runtime configuration file.

(3) FIG. **3** is a schematic of a gateway/container along with the inputs and outputs of the gateway/container.

(4) FIG. **4** is a display of the scene preprocessing performance metrics.

(5) FIG. **5** is a schematic of potential processing by a first subscriber.

(6) FIG. **6** is a flow chart showing a method of validating the authenticity of the video data throughout selection, preprocessing, publishing, accessing, and processing of the video data.

(7) FIG. **7** is a schematic of a system for blockchain recordation and validation of video data.

(8) While the above-identified figures set forth one or more embodiments of the present disclosure, other embodiments are also contemplated, as noted in the discussion. In all cases, this disclosure presents the invention by way of representation and not limitation. It should be understood that numerous other modifications and embodiments can be devised by those skilled in the art, which fall within the scope and spirit of the principles of the invention. The figures may not be drawn to scale, and applications and embodiments of the present invention may include features and components not specifically shown in the drawings.

DETAILED DESCRIPTION

(9) FIG. **1** is a schematic of system **10** for selection/extraction, preprocessing, and publishing to subscribers of video data of a region of interest (i.e., a scene) that is a subset of a first field of view of the streaming video. System **10** can include camera **12**, configuration file **14**, gateway/container **16**, and publishing location/endpoint **18**. Camera **12** can include streaming video data **20** having entire frame **22** with a first field of view. Scenes **24**A, **24**B, and **24**C (i.e., individual regions of interest) can be selected/extracted from entire frame **22** each having a second field of view, a third field of view, and a fourth field of view, respectively, that are less than the first field of view of entire frame **22**. Camera **12** collects streaming video data **20** and transfers/sends streaming video data **20** to gateway/container **16**. Gateway/container **16** preprocesses streaming video data **20** according to preprocessing parameters defined in configuration file **14** and publishes (i.e., allows access/makes available) the preprocessed video data as first video data **21**A (for scene **24**A), second video data **21**B (for scene **24**B), and third video data **21**C (for scene **24**C) to publishing location/endpoint **18**. Subscribers **26**A-**26**D can subscribe to video data **21**A-**21**A of each scene **24**A-**24**C located at endpoint **18** to access each scene **24**A-**24**C.

(10) Scene **24**A (i.e., a first region of interest) includes first video data **21**A, scene **24**B (i.e., a second region of interest) includes second video data **21**B, and scene **24**C (i.e., a third region of interest) includes third video data **21**C that are each dependent upon streaming video data **20**. In one example, first video data **21**A forming scene **24**A (i.e., a first region of interest) has a second field of view that is less than the first field of view comprising entire frame **22** shown by camera **12** (as streaming video data **20**).

(11) System **10** can include machine-readable storage media. In some examples, a machine-readable storage medium can include a non-transitory medium. The term “non-transitory” can indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In

certain examples, a non-transitory storage medium can store data that can, over time, change (e.g., in RAM or cache). In some examples, storage media can be entirely or in part a temporary memory, meaning that a primary purpose storage media is not long-term storage. Storage media, in some examples, is described as volatile memory, meaning that the memory, does not maintain stored contents when power to system **10** (or the component(s) where storage media are located) is turned off. Examples of volatile memories can include random access memories (RAM), dynamic random-access memories (DRAM), static random-access memories (SRAM), and other forms of volatile memories. In some examples, storage media can also include one or more machine-readable storage media. Storage media can be configured to store larger amounts of information than volatile memory. Storage media can further be configured for long-term storage of information. In some examples, storage media include non-volatile storage elements. Examples of such non-volatile storage elements can include magnetic hard discs, optical discs, flash memories and other forms of solid-state memory, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories. Most generally, storage media is machine-readable data storage capable of housing stored data from a stored data archive.

(12) System **10** can also include one or multiple computer/data processors. In general, the computer/data processors can include any or more than one of a processor, a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or other equivalent discrete or integrated logic circuitry. System **10** can include other components not expressly disclosed herein but that are suitable for performing the functions of system **10** and associated methods of preprocessing and processing video data and other forms of data. For example, system **10** can include communication software and/or hardware for pushing/sending configuration file **14** from storage media at a location distant from gateway/container **16**, such as cloud storage, to gateway/container **16** for execution/implementation on streaming video data **20**.

(13) Camera **12** can be any device capable of collecting streaming video data **20**, such as a Real Time Streaming Protocol (RTSP) camera or a USB camera. Streaming video data **20** can be video data that is continuously captured/recorded by camera **12** in any suitable format. Camera **12** can be positioned/located to provide streaming video data **20** displaying entire frame **22** with a first field of view. The first field of view shown/displayed by camera **12** can be a wide field of view that shows multiple regions of interest. Video data **20** being collected, for example, can show a wide field of view of a warehouse for storing commercial products and/or an assembly line producing commercial products of which each individual subscriber **26**A-**26**D may only be interested in a region/scene **24**A-**24**C that is a subset of entire frame **22**. Camera **12** can collect and transfer streaming video data **20** in any resolution/video quality and any format, including (but not limited to) MP4, AVI, FLV, WMV, MOV, MPEG, Motion JPEG, AVCHD, WebM, and/or MKV. Camera **12** can transfer/send streaming video data **20** to gateway/container **16** over any suitable means, including via the internet, short-range wireless technology, or any other type of wired and/or wireless connection.

(14) Configuration file **14** is an editable file that contains preprocessing parameters that define, among other instructions, how streaming video data **20** is to be preprocessed by gateway/container **16** to create video data **21**A-**21**C. Configuration file **14** can include numerous other instructions for gateway/container **16**, including which camera **12** to connect to (i.e., receive streaming video data **20** from), what portion of entire frame **22** to select/extract to create scenes **24**A-**24**C (i.e., how to crop frame **22** to create scenes **24**A-**24**C), and at which endpoint **18** to publish the preprocessed scenes **24**A-**24**C. This is discussed in detail with regards to FIG. **2**. Configuration file **14** can be edited/revised and pushed/conveyed to gateway/container **16** for execution in real time (i.e., runtime) such that an editor can revise the preprocessing parameters and those revisions can be applied to scenes **24**A-**24**C at runtime. Configuration file **14** can be an executable program file or have another format for including instructions and conveying information that is then used by

gateway/container **16** to apply the preprocessing to video streaming data **20**. Additionally, configuration file **14** can be stored in storage media adjacent to and/or part of gateway/container **16** or in storage media distant from gateway/container **16**, such as in the cloud. Configuration file **14** can be accessible only by one editor or can be accessible by multiple parties (e.g., editors), which may include subscribers **26**A-**26**D who can edit the preprocessing parameters defined by/within configuration file **14** to instruct gateway/container **16** to preprocess one or each of scenes **24**A-**24**C depending on the needs/desires of subscribers **26**A-**26**D, respectively.

(15) Gateway/container **16** can include a computer processor capable of performing instructions provided by configuration file **14**, which can include preprocessing parameters that are to be applied to streaming video data **20**. Gateway/container **16** can be a gateway node, edge device, container, virtual machine, or other software and/or hardware able to accept configuration file **14** and perform the instructions therein to apply the preprocessing parameters to streaming video data **20**. Further, gateway/container **16** can be within a single computer hardware set up due to virtualization. Gateway/container **16** can include one or multiple storage media for storing information, such as the preprocessing parameters pushed/sent to gateway/container **16** by/from configuration file **14** and/or other information like streaming video data **20**. Gateway/container **16** can be located at various locations, including adjacent to camera **12** and/or on the same network as camera **12**, distant from camera **12** with streaming video data **20** being received by gateway/container **16** from camera **12** via a wired or wireless connection, in the cloud, or at multiple locations. Gateway/container **16** is in communication with configuration file **14** to accept instructions for applying preprocessing parameters. Additionally, gateway/container **16** may be configured to contact configuration file **14** to determine if configuration file **14** has been edited. If edited, gateway/container **16** can perform preprocessing (on streaming video data **20** being received) according to the newly edited configuration file **14**. In other examples, gateway/container **16** can utilize preprocessing parameters or other information included within configuration file **14** on a period basis, and can utilize changes (or edits) made to configuration file **14** when encountered in a next iteration of the period.

(16) For clarity, streaming video data **20** is unaltered video data that is received by gateway/container **16** from camera **12**, whereas first video data **21**A is video data that has been preprocessed by gateway/container **16** (according to preprocessing parameters defined in configuration file **14**) to create scene **24**A. Similarly, second video data **21**B is video data that has been preprocessed to create scene **24**B, and third video data **21**C is video data that has been preprocessed to create scene **24**C. For each of first, second, and third video data **21**A-**21**C, the originating video data is streaming video data **20** (which can be converted to raw video data **20**A as described with regards to FIG. **3** below). Each of scenes **24**A-**24**C can be a subset of entire frame **22** and show second, third, and fourth fields of view, respectively, that are less than the first field of view of entire frame **22**. However, other scenes can have the same field of view as first field of view of entire frame **22** and instead, other editing is performed on that scene besides cropping; for example, that scene can be edited to be in grayscale whereas entire frame **22** of streaming video data **20** is in color.

(17) Gateway/container **16** can be in communication with endpoint **18** to which gateway/container **16** publishes the preprocessed video data **21**A-**21**C (e.g., scenes **24**A-**24**C). The communication can be wired or wireless, such as communication via the internet. However, endpoint **18** can be at the same location as gateway/container **16** or on the same computer hardware set up and/or network. Further, endpoint **18** can be located on the internet with a unique address and/or security protocol that allows for subscription and access to scenes **24**A, **24**B, and **24**C. Scenes **24**A, **24**B, and **24**C can be published to endpoint **18** using an asynchronous messaging library, for example ZeroMQ, such that scenes **24**A, **24**B, and **24**C are published as topic **1**, topic **2**, and topic **3**, respectively. Subscribers **26**A-**26**D can subscribe to any of topics **1-3** to receive video data **21**A-**21**C of scenes **24**A, **24**B, **24**C, respectively. Gateway/container **16** can publish video data **21**A-**21**C of each of

scenes **24**A-**24**C to endpoint **18** in any format suitable for use by subscribers **26**A-**26**D. For example, video data **21**A-**21**C can each be published as Motion JPEG or any of the formats listed above with regards to streaming video data **20**. The format that video data **21**A-**21**C of each of scenes **24**A-**24**C can be designated in configuration file **14** and applied to video data **21**A-**21**C by gateway/container **16**.

(18) Each scene **24**A-**24**C at topics **1-3**, respectively, can be subscribed to by any number of subscribers **26**A-**26**D. In the example shown in FIG. **1**, scene **24**A has two subscribers **26**A and **26**B, while scene **24**B has one subscriber **26**C, and scene **24**C has one subscriber **26**D. Video data **21**A-**21**C of each of scenes **24**A-**24**C can be further processed by subscriber **26**A-**26**D, respectively, depending on the desired output/inference to be determined from video data **21**A-**21**C. For example, first video data **21**A of scene **24**A can be further processed by an AI model to determine the amount of a particular product that has passed by camera **12** (in the second field of view of scene **24**A) on an assembly line. This is described in detail with regards to FIG. **5**.

(19) FIG. **2** is a schematic of configuration file **14**. Each “component” of configuration file **14** can be one or a set of instructions that, when executed by gateway/container **16**, applies a process/edit. Thus, while this disclosure may discuss the components of configuration file **14** as being physical, tangible elements, the components can be one or multiple groups of executable software code contained within configuration file **14**. Configuration file **14** includes information regarding camera credentials **28** and preprocessing parameters **30**A-**30**C corresponding to scenes **24**A-**24**C. Preprocessing parameters **30**B and **30**C can include the same type of information/parameters as preprocessing parameters **30**A. However, for simplicity, the possibilities of information/parameters shown in FIG. **2** with regards to preprocessing parameters **30**A are not shown for preprocessing parameters **30**B and **30**C. However, preprocessing parameters **30**B and **30**C can be the same or different from each other and from preprocessing parameters **30**A corresponding to scene **24**A. Preprocessing parameters **30**A can include topic name/publishing location **32**, video format **34**, accompanying information **36**, and preprocessing pipeline **38** having various video edits **38**A-**380**. Video edits **38**A-**380** can include the nonexclusive list of crop **38**A, grayscale **38**B, contrast **38**C, brightness **38**D, threshold **38**E, resize **38**F, blur **38**G, hue saturation value (HSV) **38**H, sharpen **38**I, erosion **38**J, dilation **38**K, Laplacian image processing **38**L, Sobel image processing **38**M, pyramid up **38**N, and pyramid down **380**.

(20) Configuration file **14** can be edited and pushed/conveyed to gateway/container **16** in real time (i.e., runtime) such that preprocessing parameters **30**A-**30**C (and the other information contained in configuration file **14**) can be applied to streaming video data **20** immediately to preprocess and output video data **21**A-**21**C. Configuration file **14** can be stored, edited, and/or pushed/conveyed to gateway/container **16** in any suitable format/file type, such as a text file, a comma separated value file, or other format/file type. Configuration file **14** can include other information/parameters not expressly disclosed herein and not shown in FIG. **2**.

(21) Configuration file **14** can include camera credentials **28**, which provides the information needed for gateway/container **16** to connect to camera **12** and/or receive streaming video data **20** from camera **12**. Camera credentials **28** can include other information such as encryption/decryption information, security access information, and/or instructions for beginning and/or ending the collection of streaming video data **20** by camera **12**. Camera credentials **28** can include information for connecting to multiple cameras **12** and/or information for gateway/container **16** to receive the same or different streaming video data **20** from the same or different cameras **12** for different scenes **24**A-**24**C. In one example, camera credentials **28** are provided once and applied to all scenes **24**A-**24**C. In another example, different camera credentials **28** are provided for each scene **24**A-**24**C and applied to each of scenes **24**A-**24**C individually.

(22) Configuration file **14** also includes information specific to each scene **24**A-**24**C. The information/instructions are designated as preprocessing parameters **30**A, **30**B, and **30**C, respectively. Preprocessing parameters **30**A-**30**C are used by gateway/container **16** (e.g., in the

form of executable instructions or indications of the executable instructions) and applied to streaming video data **20** to create video data **21**A-**21**C of scenes **24**A-**24**C, respectively. Preprocessing parameters **30**A-**30**C can include topic name/publishing location **32**, which designates where video data **21**A-**21**C of scenes **24**A-**24**C will be published after preprocessing. As described below with regards to gateway/container **16** and FIG. **3**, publishing location **32** can be a proxy location which is then relayed to unified endpoint **18** to make locating the topic/scenes **24**A-**24**C easier for subscribers **26**A-**26**D. Publishing location **32** can be any other location suitable for providing access to subscribers **26**A-**26**D.

(23) Configuration file **14** can designate video format **34** that each of scenes **24**A-**24**C is to be published at, which can be the same format as streaming video data **20** or any other type of suitable video format, including the formats listed above with regards to streaming video data **20** and/or video data **21**A-**21**C. Scenes **24**A-**24**C can be published having the same video format **34** or different video formats **34**. If the format of the video data of scenes **24**A-**24**C is to be changed, the steps of changing the video format can be performed before, during, or after any of the other instructions/steps set out in preprocessing parameters **30**A-**30**C. For example, video format **34** can be changed before, during, or after video edits **38**A-**380** are performed by gateway/container **16**.

(24) Preprocessing parameters **30**A-**30**C can also include accompanying information **36**, which is information provided/published with video data **21**A-**21**C for each of scenes **24**A-**24**C. Accompanying information **36** can include any information about first video data **21** that may be of use to subscribers **26**A-**26**C. For example, accompanying information **36** can include first video data **21**A frame size, which may be helpful in indicating to subscriber **26**A what processing should be performed on first video data **21**A of scene **24**A; if the frame size is 720 pixels by 486 lines, first video data **21**A of scene **24**A may be most suitable for processing by an AI model. Accompanying information **36** can include metadata and/or other information regarding what preprocessing has been performed on streaming video data **20** to create video data **21**A-**21**C for scenes **24**A-**24**C, respectively.

(25) Preprocessing parameters **30**A-**30**C can also include preprocessing pipeline **38** that includes numerous video edits **38**A-**380** that can be applied to streaming video data **20** to create video data **21**A-**21**C for each of scenes **24**A-**24**C. Preprocessing pipeline **38** can designate the instructions for the entire video edits made to streaming video data **20** for each of scenes **24**A-**24**C. The amount of time gateway/container **16** takes to perform the video edits designated by preprocessing pipeline **38** (i.e., video edits **38**A-**380**) for each of scenes **24**A-**24**C can be measured, recorded, and displayed as preprocessing performance metrics (see FIG. **4**). The order in which video edits **38**A-**380** are performed can be optimized (or otherwise enhanced for increasing accuracy, decreasing bandwidth or latency, etc.) by gateway/container **16** and/or optimized within configuration file **14** to reduce the amount of time gateway/container **16** takes to perform preprocessing pipeline **38**. This optimization can be done manually by an editor (e.g., the individual with access to edit configuration file **14**) or automatically by system **10**. For example, depending on which video edits **38**A-**380** are to be performed in preprocessing pipeline **38** by gateway/container **16**, the order of performance of those video edits **38**A-**380** can be rearranged to reduce the amount of time gateway/container **16** takes to perform preprocessing pipeline **38**. In one example, crop **38**A is the first video edit **38**A-**380** to be performed, followed by other video edits **38**B-**380**.

(26) Video edits **38**A-**380** are a nonexclusive list of edits that can be designated in configuration file **14** and performed on streaming video data **20** by gateway/container **16**. Preprocessing pipeline **38** can include other video edits not expressly included in the list of video edits **38**A-**380**. Similarly, not all of video edits **38**A-**380** need to be performed to create video data **21**A-**21**C of each of scenes **24**A-**24**C, and different scenes **24**A-**24**C can include different video edits **38**A-**380** performed on streaming video data **20** by gateway/container **16**. In one example, only crop **38**A is performed on streaming video data **20** to create first video data **21**A of scene **24**A, while a different crop **38**A and brightness **38**D, resize **38**E, and dilation **38**K are performed on streaming video data **20** to create

second video data **21**B of scene **24**B that is different than first video data **21**A of scene **24**A.

(27) Each of video edits **38**A-**380** are briefly described as follows. Crop **38**A is the removal of unnecessary areas/regions (i.e., regions that are not of-interest to the subscriber) of entire frame **22** having first field of view to create scenes **24**A-**24**C each with second, third, and fourth field of views, respectively. Scenes **24**A-**24**C that have been cropped **38**A have fields of view that are a subset of (i.e., less than) first field of view of entire frame **22**. Grayscale **38**B is the alteration of the color of video data **21**A-**21**C and can include limiting the color to be between white and black. Contrast **38**C is the alteration of the difference between the maximum and minimum pixel intensity. Brightness **38**D is the alteration in the luminous brightness of video data **21**A-**21**C. Threshold **38**E is the alteration of the color of video data **21**A-**21**C by changing the color of select pixels of video data **21**A-**21**C that are above a specified threshold color value. Resize **38**F is the alteration of the frame size of video data **21**A-**21**C without cutting/cropping any of the frame out. Blur **38**G is the alteration of the clarity of video data **21**A-**21**C, which may be desired for some processing applications, such as an AI model, performed by subscribers **26**A-**26**D. Hue saturation value (HSV) **38**H is assigning a numerical readout of video data **21**A-**21**C that corresponds to the color contained therein. Sharpen **38**I is altering video data **21**A-**21**C to make the objects therein appear more defined/sharpened. Erosion **38**J is altering video data **21**A-**21**C by shrinking pixels and/or removing pixels on object boundaries, while dilution **38**K is the reverse of erosion in that video data **21**A-**21**C is enlarged by resizing pixels and/or adding pixels at object boundaries. Laplacian image processing **38**L and Sobel image processing **38**M are processing techniques known in the art that can be applied to video data **21**A-**21**C. Pyramid up **38**N and pyramid down **380** are altering video data **21**A-**21**C by smoothing and/or subsampling as known in the art. Each of scenes **24**A-**24**C can include these and other video edits **38**A-**380** to be applied by gateway/container **16** to preprocess streaming video data **20** and output as scenes **24**A-**24**C for use by subscribers **26**A-**26**D.

(28) Configuration file **14** can arrange the instructions of camera credentials **28** and preprocessing parameters **30**A-**30**C to be performed in any order, or gateway/container **16** can have the capabilities to arrange/rearrange the information/instructions to be performed in a desired/optimized sequence. Additionally, gateway/container **16** can be configured to perform each set of preprocessing parameters **30**A-**30**C in parallel such that preprocessing parameters **30**A, preprocessing parameters **30**B, and preprocessing parameters **30**C are performed at the same time (and the time gateway/container **16** takes to perform those preprocessing parameters are measured, recorded, and displayed as metrics). Configuration file **14** can be edited at any time by an editor and then pushed/conveyed/accessed by gateway/container **16** at runtime such that the preprocessing of video data **21**A-**21**C is altered according to the newly edited configuration file **14** at runtime.

(29) FIG. **3** is a schematic of gateway/container **16** (hereinafter, "gateway **16**") along with inputs to and outputs from gateway **16**. Each "component" of gateway **16** (and corresponding inputs and outputs) can be one or a set of instructions, programs, processors, storage media locations, and/or other software or hardware used to select/extract, preprocess, and publish video data **21**A-**21**C as scenes **24**A-**24**C. Thus, while this disclosure may discuss the components of gateway **16** (and corresponding inputs and output) as being physical, tangible elements, the components can be partially or entirely contained within software and/or hardware.

(30) Inputs to gateway **16** can be streaming video data **20** (received from camera **12**) and configuration file **14**, which includes camera credentials **28** and preprocessing parameters **30**A-**30**C. Outputs from gateway **16** can be scenes **24**A-**24**C to unified endpoint **18**, which is the location at which scenes **24**A-**24**C are published as topics **1-3**, and metrics **40**, which is the location at which preprocessing pipeline **38** information (i.e., the amount of time gateway **16** takes to apply preprocessing parameters **30**A-**30**C to streaming video data **20** to create scenes **24**A-**24**C) is published/accessible. Gateway **16** can include raw video data **20**A, which is streaming video data **20** that has been published at internal topic **42** and to which gateway **16** subscribes to receive video data used to create each of scenes **24**A-**24**C. Gateway **16** can include preprocessing pipeline

optimization and preprocessing **44**, which uses preprocessing parameters **30**A-**30**C as defined in configuration file **14** to preprocess streaming video data **20** (accessed as raw video data **20**A at internal topic **42**) to create scenes **24**A-**24**C. Gateway **16** publishes scenes **24**A-**24**C to proxy location **46** at topics **1-3**, respectively. Gateway **16** can then publish/relay scenes **24**A-**24**C (having video data **21**A-**21**C) from proxy location **46** to unified endpoint **18**.

(31) Gateway **16** receives streaming video data **20** from camera **12** and can publish the video data as raw video data **20**A at internal topic **42**. This configuration provides for a constant, known location of published raw video data **20**A independent of where the original streaming video data **20** is received from. Thus, if the location where streaming video data **20** is being received from changes (e.g., if one camera is disconnected but another camera comes online), raw video **20**A will still be accessible at internal topic **42** without the need to change where gateway **16** is looking for video data to create scenes **24**A-**24**C, thus ensuring a smooth transition of incoming streaming video data **20**. Raw video data **20**A published at internal topic **42** can be configured such that only gateway **16** has access.

(32) Gateway **16** can be provided with the information in configuration file **14** (i.e., camera credentials **28** and preprocessing parameters **30**A-**30**C) via a variety of avenues. In one example, gateway **16** has location information of configuration file **14** and actively accesses configuration file **14**. In another example, configuration file **14** is pushed/conveyed to gateway **16** once, periodically, or continuously and gateway **16** passively waits to begin preprocessing streaming video data **20** until configuration file **14** has been received. Another example can be a combination of the two above examples in that gateway **16** actively accesses configuration file **14** at the beginning of preprocessing (and continues preprocessing following those known instructions) and configuration file **14** is pushed/conveyed to gateway **16** only after configuration file **14** has been edited/revised.

(33) Gateway **16** can perform pipeline optimization on preprocessing parameters **30**A-**30**C As discussed above, pipeline optimization can be performed by gateway **16** (or another component) to reduce the time gateway **16** takes to preprocess raw video data **20**A to create video data **21**A-**21**C (and attach any additional information) of scenes **24**A-**24**C and/or to increase accuracy of the preprocessing operations. Pipeline optimization can include arranging/rearranging the order in which video edits **38**A-**380** are performed by gateway **16**.

(34) Gateway **16** can then, according to preprocessing parameters **30**A-**30**C defined within configuration file **14**, preprocess **44** raw video data **20**A (which is derived from and similar to streaming video data **20**) to edit raw video data **20**A to create video data **21**A-**21**C of scenes **24**A-**24**C. The preprocessing to create each of scenes **24**A-**24**C can be performed individually for each scene and can be performed in parallel (i.e., simultaneously). The preprocessing performed by gateway **16** can edit each of scenes **24**A-**24**C to the desires/needs of subscribers **26**A-**26**C, respectively. For example, gateway **16** can preprocess raw video data **20**A to crop a first field of view of entire frame **22** to eliminate areas/regions of the first field of view that are of no interest to subscriber **26**A and keep a second field of view of, for example, scene **24**A that is a subset of the first field of view. Thus, further processing by subscriber **26**A (or subscriber **26**B) does not need to be performed on the areas (regions not of-interest) eliminated/trimmed by the cropping performed during preprocessing. In this example, the cropping reduces the processing time and resources needed by subscriber **26**A. The alteration of video format can be included in the preprocessing and/or can be performed before, during, or after the application of other preprocessing parameters **30**A-**30**C.

(35) Scenes **24**A-**24**C, which are made up of raw video data **20**A that has been preprocessed according to configuration file **14** to create video data **21**A-**21**C, are published/sent to proxy location **44**. Because scenes **24**A-**24**C are continuous video data, scenes **24**A-**24**C are continuously published (e.g., made available) to subscribers **26**A-**26**C, respectively. Scenes **24**A-**24**C are published/sent first to proxy location **44** and then relayed to unified endpoint **18**. Having scenes

24A-**24**C first being published/sent to proxy location **44** ensures that, no matter what path scenes **24**A-**24**C take, video data **21**A-**21**C of scenes **24**A-**24**C will always end up at proxy location **44** and then be relayed to unified endpoint **18**. Because scenes **24**A-**24**C always end up at proxy location **44**, unified endpoint **18** always knows the location to access scenes **24**A-**24**C and can actively look to proxy location **44** to obtain scenes **24**A-**24**C or passively wait for scenes **24**A-**24**C to be published/sent to unified endpoint **18**.
(36) During preprocessing to create scenes **24**A-**24**C, the amount of time gateway **16** takes to apply all of preprocessing parameters **30**A-**30**C to raw video data **20**A to create scenes **24**A-**24**C, respectively, (or the amount of time gateway **16** takes to apply only video edits **38**A-**380** of preprocessing pipeline **38**, depending on the desired measurement) is measured, recorded, and published at metrics **40** for viewing by an editor or any party with access. Metrics **40** can be published on an asynchronous messaging library like ZeroMQ (similar to the publishing of scenes **24**A-**24**C) or can be displayed on a user interface similar to FIG. **4**, which shows a display of scene preprocessing performance metrics **40**A. Metrics **40** can be outputted from gateway **16** to numerous other systems and/or locations, including to the cloud or another location distant from gateway **16**.
(37) As shown in FIG. **4**, scene preprocessing performance metrics **40**A show a value of time for each of scenes **24**A-**24**C. This value of time displayed is the amount of time gateway **16** takes to apply preprocessing parameters **30**A-**30**C for each of scenes **24**A-**24**C. Alternatively, this value of time displayed can be the amount of time gateway **16** takes to apply only video edits **38**A-**380** of preprocessing pipeline **38** and not the entirety of preprocessing parameters **30**A-**30**C. This measurement may be desired because the other instructions/information of preprocessing parameters **30**A-**30**C (other than video edits **38**A-**380**) may not change between scenes **24**A-**24**C while preprocessing pipeline **38** (the video edits **38**A-**380** being applied) may be drastically different from scene to scene and thus the amount of time for performing preprocessing pipeline **38** may vary greatly from scene to scene.
(38) For example, preprocessing performance metrics **40**A in FIG. **4** show an amount of time gateway **16** takes to perform the video edits of preprocessing pipeline **38** for scene **24**A as 5.2 milliseconds (ms), for scene **24**B as 9.4 ms, and for scene **24**C as 3.3 ms. In this example, it may be desirable or necessary to alter preprocessing parameters **30**B in configuration file **14** (either remove some edit and/or rearrange the order in which those edits are performed by gateway **16**) to reduce the amount of time gateway **16** takes to apply preprocessing pipeline **38** to create/alter scene **24**B. As discussed above, gateway **16** can have the capability to optimize the order in which the edits/instructions in preprocessing parameters **30**A-**30**C and/or preprocessing pipelines **38** are performed to reduce the amount of time needed to preprocess/apply the edits/instructions. Thus, preprocessing performance metrics **40**A as shown in FIG. **4** may be the shortest amount of time gateway **16** takes to perform preprocessing parameters **30**A-**30**C (or preprocessing pipeline **38**, depending on the chosen measuring points).
(39) Scenes **24**A-**24**C can be published from proxy location **44** to unified endpoint **18** as topics **1-3** within an asynchronous messaging library, such as ZeroMQ. Unified endpoint **18** allows for a known, constant location to which subscribers **26**A-**26**C can look to access scenes **24**A-**24**C, respectively. If more scenes are created, those scenes would be published and accessible at unified endpoint **18**, so subscribers **26**A-**26**D and/or other subscribers would know where to look to access the additional scenes.
(40) FIG. **5** is a schematic showing processing **48** capable of being performed on scene **24**A by first subscriber **26**A. Processing **48** can also be performed on scenes **24**B and **24**C or other scenes not disclosed herein.
(41) After being published/sent to endpoint **18**, scenes **24**A-**24**C are available to be subscribed to and accessed by subscribers **26**A-**26**D. Each scene can be subscribed to by any number of subscribers as is necessary/desirable. For example, a scene may be subscribed to by numerous subscribers each running processing **48** that includes different AI models. In this example, one AI

model can be determining the amount of a first product that is passing through the scene on an assembly line, while a second AI model can be determining the amount of a second product that is passing through the scene on the same assembly line. In this case, the scene is unchanged between the two AI models (i.e., between the two subscribers) but the processing performed by each subscriber after the scene has been published is different. It should be noted that one entity (e.g., a person, company, quality control sector) can subscribe to a scene multiple times and thus constitute multiple subscribers. As discussed above, the preprocessing performed to create/alter each scene can be tailored to the needs of the subscriber(s) to, for example, reduce processing **48** resources and time needed to determine at least one output that is indicative of an inference the subscriber is aiming to ascertain.

(42) The example in FIG. **5** shows first subscriber **26**A performing processing **48** on scene **24**A. Processing **48**, as selected and executed by subscriber **26**A (either automatically and/or manually by a computer processor and/or other hardware and software), can include AI model **48**A, optical character recognition (OCR) **48**B, video clipping **48**C, further formatting **48**D, and display **48**E of the video data of scene **24**A. Processing **48** can include other instructions/edits not expressly disclosed in FIG. **5** and listed above.

(43) The disclosed potential instructions/edits that subscriber **26**A can perform in processing **48** are as follows. AI model **48**A can be a program/model that may have machine learning and can use scene **24**A to determine at least one output indicative of an inference dependent upon scene **24**A. The inference, for example, can be the amount of a specific product that is viewable in scene **24**A over a defined period of time. AI model **48**A can also be, for example, a program/model that determines how many people appear in scene **24**A over a defined period of time. AI model **48**A can include other capabilities and/or configurations. OCR **48**B can be a program (or other configuration) that recognizes and records any characters (i.e., text) that appear in scene **24**A. For example, scene **24**A can be video data of a street and OCR **48**B will recognize and record any text that appears on the side of a vehicle, such as a delivery truck, that is in scene **24**A. Video clipping **48**C can clip the video data shown in scene **24**A to create a clip of a defined period of time, and/or video clipping **48**C can clip the video data shown in scene **24**A to create a static image of a defined moment in time. Further formatting **48**D can be video edits, such as video edits **38**A-**380** in configuration file **14**, or any other video or file formatting that are performed by subscriber **26**A. For example, further formatting **48**D can include cropping scene **24**A to be a subset of the second field of view shown in scene **24**A. Display **48**E can be making scene **24**A viewable on a screen or other visual display. Display **48**E can also include any video formatting/reconfiguring that is necessary to effectuate the display of scene **24**A. While video edits **38**A-**380** and further formatting **48**D to create and/or process video data **21**A-**21**C have included only edits to the video/image, edits can include editing the audio or other aspects of the video data.

(44) The potential instructions/edits (i.e., processing **48**) can be performed in parallel or series. Further, processing **48** can be configured such that instructions/edits **48**A-**48**E work together such that one instruction/edit is prompted by an inference from another instruction/edit. For example, video clipping **48**C can be configured to work in tandem with another edit/process; if AI model **48**A determines that a product is defective, video clipping **48**C can be prompted to record and clip a particular duration (or moment to create a static image) of scene **24**A showing the defective product and save the clip (or image) for proof/validation.

(45) System **10**, with associated methods, for selection/extraction, preprocessing, and publishing of streaming video data **20** into scenes **24**A-**24**C and for processing scenes **24**A-**24**C is used to determine at least one output that is indicative of an inference dependent upon video data **21**A-**21**C. System **10** reduces the preprocessing/processing time and resources necessary for accomplishing the desired output/determining the selected inference. System **10** allows for a streamlined process that extends from collecting streaming video data **20** from camera **12** to making preprocessing scenes **24**A-**24**C available to subscribers **26**A-**26**C for further processing and

analysis/determinations. System **10** also allows for measurement, recordation, and viewing of preprocessing performance metrics **40** and optimization of preprocessing pipeline **38** (and/or preprocessing parameters **30**A-**30**C) to reduce the amount of time and resources needed to apply preprocessing parameters **30**A-**30**C to streaming video data **20** (i.e., raw video data **20**A) to create video data **21**A-**21**C of scenes **24**A-**24**C.

(46) FIG. **6** shows a flow chart of method **50** for validating the authenticity of video data throughout selection, preprocessing, publishing, accessing, and processing of the video data. Throughout method **50**, the video data after each step is committed to a blockchain ledger to record and authenticate the video data to ensure the video data has not been manipulated and to keep a record of the state/properties of the video data after receiving, preprocessing, publishing, accessing, and processing the video data. Committing the video data to the blockchain ledger, with validation by multiple nodes of the distributed ledger, after each step and/or after the video data has changed hands (from one computer processor, network, and/or location to another) prevents manipulation of the video data without evidence of the manipulation because the video data committed to the blockchain ledger cannot be edited/modified/manipulated after the video data has been committed to the blockchain ledger. Thus, if the video data has been manipulated, a clear record has been formed and examination of the blockchain entries of the video data (and associated metadata) will show when (and by whom) the video data was manipulated.

(47) Method **50** includes many of the same steps as set out in the method associated with system **10**. However, method **50** includes the steps of committing the video data (and associated metadata) to the blockchain ledger regularly. Method **50** can include other steps not expressly disclosed herein, and the steps of method **50** can be altered and/or performed in any sequence without departing from the scope of the invention.

(48) Step **52** includes collecting streaming video data by a camera or other streaming video source. The video data can be collected by, for example, camera **12**. The collection of the streaming video data can be the creation of the video data by continuously recording a potentially ever-changing frame (with a field of view) by the camera. For example, camera **12** can record the inside of a warehouse in which consumer products are temporarily stored before being shipped out to one or multiple stores. The collection of streaming video data can be the recording of the movement of those products (and machinery used to move the products).

(49) Step **54** includes committing the streaming video data and associated metadata to a blockchain ledger. Step **54** can be performed after the streaming video data (and associated metadata) has been collected/created, or step **54** can be performed at a time before the video data is conveyed/sent to a first computer processor (e.g., gateway **16**). Alternatively, step **54** of committing the video data and metadata to the blockchain ledger can be performed at any other time before being conveyed/sent out from the camera. With the streaming video data and metadata being committed to the blockchain ledger prior to transmission of the video data and metadata from the camera (or from the network to which the camera is connected), a first chain/record of the video data and metadata is recorded in the blockchain ledger for later validation and authentication.

(50) Committing the streaming video data (or first video data or altered first video data) to the blockchain ledger can include submitting and saving the streaming video data (and associated metadata) in multiple nodes of a distributed ledger (i.e., distributed ledgers). The distributed ledgers interact to confirm that the streaming video data submitted and saved is identical across all ledgers. Thus, the video data cannot practically be changed/manipulated after it has been committed to the blockchain ledger without evidence of such manipulation because the changes would need to be made to all of the identical entries of that streaming video data in each of the distributed ledgers. This characteristic of the blockchain ledger allows for commitment/recordation of the video data to prevent manipulations/modifications of the video data.

(51) Step **56** includes conveying/sending the streaming video data and metadata from the camera to the first computer processor, such as gateway **16**. In other words, step **56** includes receiving the

streaming video data and metadata by the first computer processor (e.g., gateway **16**). As with system **10** above, camera **12** and gateway **16** can be, for example, distant from one another or physically or virtually adjacent to one another, such as being on the same network.

(52) Step **58** includes committing the streaming video data and metadata to the blockchain ledger. After the streaming video data (and metadata) has been transmitted from the camera to the first computer processor (e.g., gateway **16**), the streaming video data and metadata can be committed to the blockchain ledger to ensure that the streaming video data and metadata has not been manipulated between the time the streaming video data was collected by the camera and the time the streaming video data is received by the first computer processor. Manipulation of the video data can be identified, for example, by comparing the streaming video data and metadata recorded in the blockchain ledger in step **54** to the streaming video data and metadata recording in the blockchain ledger in step **58**. If the two entries of streaming video data (and metadata) are different, then the streaming video data (and metadata) can be identified as altered or otherwise manipulated. If the two entries are identical, then the streaming video data (and metadata) can be identified as having not been altered or otherwise manipulated.

(53) Step **60** includes preprocessing the streaming/incoming video data, by the first computer processor, according to preprocessing parameters defined within the configuration file (which can be edited/revised and updated at runtime). The preprocessing can include formatting the streaming/incoming video data to create first video data that is a subset of the streaming/incoming video data. For example, the first video data can have a first field of view that is less than a second field of view of the streaming/incoming video data collected by the camera. The preprocessing of the streaming video data according to the preprocessing parameters in the runtime configuration file to create the first video data can include other edits to the video data and the associated metadata as described above with regards to FIGS. **1-3**.

(54) Step **62** includes committing the first video data and metadata to the blockchain ledger. After the first video data (and associated metadata) is preprocessed, the first video data and metadata are committed to the blockchain ledger to record the edits that have been performed on the first video data (and metadata) by the first computer processor (e.g., gateway **16**) according to the preprocessing parameters defined in the configuration file. Step **62**, in committing the first video data and metadata to the blockchain ledger after the first video data has been preprocessed, ensures that if additional edits (e.g., manipulations) to the first video data (other than those performed prior to step **62**) are later found when examining the first video data, it can determined that those edits/manipulations were not performed by the first computer processor (e.g., gateway **16**) or at the time that the first video data was committed to the blockchain ledger.

(55) Step **64** includes publishing the first video data (and associated metadata) to the endpoint, which can be a proxy location and/or unified endpoint hosted by the first computer processor (e.g., gateway **16**). The publishing in step **64** can be to/within an asynchronous messaging library, such as ZeroMQ. Further, the proxy location and/or unified endpoint to which the first video data (e.g., scenes **24**A-**24**C) is published can be in wired or wireless communication with the first computer processor (e.g., gateway **16**).

(56) Step **66** includes accessing the first video data (e.g., the scene/topic) and associated metadata by the first subscriber (or any number of subscribers). The first subscriber can be a second computer processor or any other type of software and/or hardware. The preprocessed first video data can be edited/preprocessed for a desired use by the first subscriber, who then accesses the first video data (and associated metadata) potentially for further processing depending on the desired output/inference to be determined from the first video data. The first subscriber can be distant (e.g., physically remote) from the location at which the first video data is published such that step **66** can be performed physically and temporarily distant from step **64**. For example, the first subscriber can be a second computer processor that is in communication with the endpoint via the internet, with the first video data being published to the endpoint by the first computer processor.

(57) Step **68** includes again committing the first video data and metadata to the blockchain ledger. After the first video data (and metadata) has been transmitted from the first computer processor (e.g., gateway **16**) to the first subscriber (e.g., the second computer processor and/or subscribers **26**A-**26**D), the first video data and metadata can be committed to the blockchain ledger to ensure that the first video data and metadata has not been manipulated between the time the first video data was published by the first computer processor and the time the first video data is received by the first subscriber. Manipulation can be identified by comparing the first video data and metadata recorded in the blockchain ledger in step **62** to the first video data and metadata recording in the blockchain ledger in step **68**. If the two entries of the first video data (and metadata) are different, then the first video data and metadata can be identified as altered/manipulated. If the two entries are identical, then the first video data and metadata can be identified as having not been altered/manipulated.
(58) Step **70** includes processing the first video data, by the first subscriber (e.g., the second computer processor), to determine at least one output indicative of an inference dependent upon the first video data and/or to alter the first video data. For simplicity, the first video data (and associated metadata) that has been processed in step **70** can be referred to as altered first video data because the first video data can be changed/altered by the processing performed in step **70**. Examples of the processing by the first subscriber are described above with regards to FIG. **5**. Step **70** can include further editing of the first video data and/or metadata, which may change/alter the first video data as compared to the first video data before step **70** (i.e., altered first video data). Additionally, the processing of the first video data in step **70** can include, for example, applying an AI model to the first video data to determine and record at least one output. In another example, optical character recognition (OCR) is applied to the first video data to determine and record any text that is shown in the first video data. Step **70** can include other processing to extract information/data from the first video data in the form of one or more outputs.
(59) Step **72** includes committing the altered first video data and metadata to the blockchain ledger. After the altered first video data (and associated metadata) is processed by the first subscriber (e.g., the second computer processor), the altered first video data and metadata are committed to the blockchain ledger to record the edits/processing that have been performed on the first video data (and metadata) by the first subscriber (e.g., the second computer processor). Step **72**, in committing the altered first video data and metadata to the blockchain ledger after the first video data has been processed, ensures that if additional edits (e.g., manipulations) to the altered first video data (other than those performed prior to step **72**) are later identified when examining the altered first video data, it can be determined that those edits/manipulations were not performed by the first subscriber (e.g., the second computer processor).
(60) Step **72** can also include committing any information recorded during step **70** (during the processing by the first subscriber) to the blockchain ledger. This information can include, for example, any information/data created/collected by an AI model (such as the amount of a product that passes by in the first video data) and/or any text collected by OCR. Committing the additional information created/collected/recorded during step **70** provides a record of the information/data that cannot be altered/modified once committed to the blockchain ledger. Such information/data may be used for proof of the particular output determined by the processing, as well as validation and/or authentication of the information and/or inferences derived from the information.
(61) Step **74**, which can be performed at any point during method **50**, includes examining the streaming video data, the first video data, and/or the altered video data and associated metadata as committed/recorded in the blockchain ledger. Because the video data recorded in the blockchain ledger is not able to be modified/altered, the video data is authentic and used as proof of the state of the video data throughout the collection, preprocessing, publishing, accessing, and processing of that video data. If the video data was manipulated at any point during method **50** (without notice and/or authorization), the manipulated video data would be recorded as is in the blockchain ledger

and/or comparable to a previous entry of the video data in the blockchain ledger. Both situations, after step **74** of examining the video data (and associated metadata) in the blockchain ledger, allow for an identification of when and who manipulated the video data without notice and/or authorization. Thus, method **50**, with multiple steps that include committing the video data to the blockchain ledger, allows for validation of the authenticity of the video data throughout the collection, preprocessing, publishing, accessing, and processing of the video data.

(62) Step **74** can include comparing the streaming video data (and associated metadata) as committed to the blockchain ledger in step **54** and/or step **58** to the first video data (and associated metadata) as committed to the blockchain ledger in step **62** and/or step **68** and validating that the only difference between the streaming video data and the first video data is the changes made during step **60**, which is preprocessing the steaming video data. Additionally, step **74** can include comparing the first video data (and associated metadata) as committed to the blockchain ledger in step **62** and/or step **68** to the altered first video data (and associated metadata) as committed to the blockchain ledger in step **72** and validating that the only difference between the first video data and the altered video data is the changes made during step **70**, which is processing the first video data. Moreover, step **74** can compare the streaming video data that was committed to the blockchain ledger to the altered first video data that was committed to the blockchain ledger in a similar fashion to examine the differences and validate that the only difference is the changes made to the streaming video data during step **60** and step **70**. If the differences between the video data entries into the blockchain ledger are other than the changes performed during step **60** and **70**, it can be determined that the video data was manipulated/altered/modified without notice and/or authorization during method **50**. In such a situation, the video data would not be validated as authentic.

(63) FIG. **7** shows a system for blockchain recordation (i.e., commitment) and authentication of video data. Unified endpoint **18** provides scene **24**A (topic **1**) having first video data **21** to blockchain validation system **110** via providing availability of scene **24**A via publication at unified endpoint **18** (similar to system **10**). To access scene **24**A (video data **21**), first AI model subscriber **126**A, second AI model subscriber **126**B, and blockchain subscriber **126**C subscribe to topic **1** at unified endpoint **18**. Each of first AI model subscriber **126**A, second AI model subscriber **126**B, and/or blockchain subscriber **126**C can be one or multiple computer processors at various locations near or distant from unified endpoint **18** and each other. First AI model subscriber **126**A, second AI model subscriber **126**B, and/or blockchain subscriber **126**C can be in communication with sink **176**, distributed ledger **178** (also referred to as blockchain ledger **178**), and/or database **180**.

(64) First AI model subscriber **126**A performs processing on first video data **21** that is similar to (or the same as) AI model **48**A, which is one of processing **48** that is performed on first video data **21**. Additionally, second AI model subscriber **126**B performs processing on first video data **21** that is similar to (or the same as) AI model **48**A, but in system **110** first AI model subscriber **126**A and second AI model **126**B can be analyzing/processing first video data **21** to determine outputs that are indicative of different inferences. The determination of an output that is indicative of an inference can include monitoring, by at least one of AI model subscribers **126**A and **126**B, video data **21** for the occurrence of the inference and formulating the first output when the inference occurs. Thus, the output is dependent upon the occurrence of an inference.

(65) When an inference is detected/determined, AI model subscribers **126**A and/or **126**B can formulate the corresponding outputs (e.g., the detection of an inference prompts the formulation of an output). For example, first AI model **126**A can be processing first video data **21** to determine an output that is indicative of the amount of a specific product that is viewable in scene **24**A (i.e., a first inference), and second AI model **126**B can be processing first video data **21** to determine a different output that is indicative of how many people appear in scene **24**A over a defined period of time (i.e., a second inference). The outputs by first AI model subscriber **126**A and second AI model subscriber **126**B can be annotated first video data and metadata (shown in FIG. **7** as first annotated

video data **121**A and second annotated video data **121**B) that includes information detailing when, where, and why the inferences were determined by the AI models. This information can be included by annotating the video data and/or adding information to the metadata associated with the annotated video data.

(66) For example, first annotated video data **121**A can show an outlined box overlaid over original video data **21** highlighting a specific product that is viewable in scene **24**A as well as a number overlaid over original video data **21** showing a probability that first AI model subscriber **126**A has correctly identified the specific product. The metadata associated with first annotated video data **121**A can include, in some examples, additional information related to the annotated video data, such as specifying the location on the frame of the video at which the specific product is viewable. Annotated video data **121**A and **121**B (and associated metadata) can include any other information used by and/or created by AI model subscribers **126**A and **126**B, respectively.

(67) First AI model subscriber **126**A and second AI model subscriber **126**B can process video data **21** in parallel (as shown in FIG. **7**) where each subscriber **126**A and **126**B subscribes to topic **1** to access video data **21** and each subscriber **126**A and **126**B performs processing/analysis on original video data **21** simultaneously to determine the occurrence of first inferences and second interferences associated with first AI model subscriber **126**A and second AI model subscriber **126**B, respectively. In another embodiment, first AI model subscriber **126**A and second AI model subscriber **126**B perform processing on video data **21** in series such that first AI model subscriber **126**A has access to video data **21** and processes video data **21** first and second AI model subscriber **126**B has access to annotated video data **121**A and processes annotated video data **121**A after it has been processed by first AI model subscriber **126**A, thereby outputting the information from both first AI model subscriber **126**A and second AI model subscriber **126**B as second annotated video data **121**B. While shown in FIG. **7** as only having two AI model subscribers **126**A and **126**B performing processing in parallel, other embodiments of system **110** can include any number of AI model subscribers performing processing in series and/or in parallel.

(68) Blockchain subscriber **126**C accesses video data **21** by subscribing to topic **1** (i.e., scene **24**A). Blockchain subscriber **126**C can be configured to perform no processing of video data **21** and instead can just provide a relay (e.g., location) for original video data **21** for later use in commitment to distributed ledger **178** and database **180** along with first annotated video data **121**A and/or second annotated video data **121**B. Original video data **21** as accessed by blockchain subscriber **126**C may be useful in comparison to first annotated video data **121**A (which includes information added by first AI model subscriber **126**A) and/or second annotated video data **121**B (which includes information added by second AI model subscriber **126**B). System **110** can be configured to not include blockchain subscriber **126**C. For example, sink **176** could be configured to perform the functions of blockchain subscriber **126**C by subscribing to topic **1** at unified endpoint **18**. Additionally, blockchain subscriber **126**C can be combined with one or more other components of system **110**, such as being a part of sink **176**.

(69) Sink **176** can include one or more controllers, computer processors, digital storage media, and/or other components that are in communication with first AI model subscriber **126**A, second AI model subscriber **126**B, and blockchain subscriber **126**C. Sink **176** can accept and store first annotated video data **121**A (and associated metadata), second annotated video data **121**B (and associated metadata), and/or original first video data **21** and commit video data **121**A, **121**B, and/or **21** (and associated metadata) to distributed ledger **178** and/or database **180** when a trigger event occurs, which can be dependent upon the outputs/inferences determined by first AI model subscriber **126**A and/or second AI model subscriber **126**B. Alternatively, sink **176** does not need to store video data **121**A, **121**B, and/or **21** (and associated metadata) and can be a component that, when a trigger event occurs, performs instructions to commit video data **121**A, **121**B, and/or **21** (and associated metadata) to distributed ledger **178** and/or database **180**. Additionally, sink **176** can be a virtual location to which video data **121**A, **121**B, and **21** is relayed/sent/published until a

trigger event prompts commitment to distributed ledger **178** and/or database **180**

(70) The event that triggers the commitment of video data **121**A, **121**B, and/or **21** to distributed ledger **178** and/or database **180** can be any event for which recordation and authentication is desired (e.g., proof of authenticity and/or that AI model subscribers **126**A and/or **126**B are functioning properly). The trigger event can be an instance (one frame of video data or a duration of time encompassing multiple frames of video data) when AI model subscribers **126**A and **126**B determine an output indicative of an inference. The inference can be anything of interest to AI model subscribers **126**A and/or **126**B. Some examples of an inference are as follows: a product that is viewable in video data **21** (i.e., scene **24**A), an incorrect product that is viewable in video data **21** (e.g., a product that is not the product of interest), a product that is viewable in video data **21** that is defective, a specified number of products that are viewable in video data **21** simultaneously or over a specified duration of time, any person that is viewable in video data **21**, any vehicles that are viewable in video data **21**, or anything else that is of interest to AI model subscribers **126**A and/or **126**B. In another example, video data **21** can show at least a portion of a manufacturing system used to construct a specific product and the inference occurs when the AI model determines that the specific product is incorrectly constructed (i.e., has a defect).

(71) For ease of understanding, each commitment of video data **121**A, **121**B, and/or **21** (and associated metadata, which can include customer, business, and/or other related data) to distributed ledger **178** and/or database **180** is called an entry. Sink **176** can review video data **121**A, **121**B, and/or **21** for outputs indicative of inferences that are trigger events, or every inference determined by AI model subscribers **126**A and **126**B can be a trigger event that results in an entry being committed to blockchain ledger **178** and database **180**. Other embodiments/configurations can have first AI model subscriber **126**A, second AI model subscriber **126**B, and/or blockchain subscriber **126**C in communication with one another and configured to instruct one another to commit an entry to distributed ledger **178** and/or database **180** when each determines an output indicative of an inference that is a trigger event. In such a configuration, each subscriber **126**A, **126**B, and **126**C determines when one or multiple subscribers should provide video data to be part of an entry committed to distributed ledger **178** and/or database **180**. Further, system **110** can be configured such that the presence/determination of an inference by AI model subscribers **126**A and/or **126**B automatically triggers the commitment of an entry that encompasses that inference (along with original video data **21**) without the need for further instruction from sink **176** or other components.

(72) Each entry to distributed ledger **178** and/or database **180** can include video data from one, two, or all of video data **121**A, **121**B, and/or **21** (and associated metadata). For example, when first AI model subscriber **126**A determines an output that is indicative of an inference that warrants the commitment of an entry, both first annotated video data **121**A (from first AI model subscriber **126**A) and original video data **21** (from blockchain subscriber **126**C) can be committed as an entry to distributed ledger **178** and/or database **180** (without the committing of second annotated video data **121**B). In another situation, when first AI model subscriber **126**A determines an output that is indicative of an inference that warrants the commitment of an entry, all three of first annotated video data **121**A, second annotated video data **121**B, and original video data **21** can be committed as an entry to distributed ledger **178** and/or database **180**.

(73) Each entry to distributed ledger **178** and/or database **180** can be one frame of specified video data (e.g., the frame/still image of video data that shows the inference determined by first AI model subscriber **126**A and/or second IA model subscriber **126**B as well as the one frame of original video data **21** of blockchain subscriber **126**C) or multiple frames/still images of the video data. For example, if first AI model subscriber **126**A determines an output that is indicative of an inference warranting commitment of an entry to distributed ledger **178** and/or database **180**, the entry can include ten frames of video data before the frame of interest (i.e., the frame that shows the inference), ten frames of video data after the frame of interest, and the frame of interest. Thus, the entry would include twenty-one frames of first annotated video data **121**A. Additionally, the entry

could also include those same corresponding twenty-one frames of original video data **21** for comparison to first annotated video data **121**A. In such a configuration, with each entry including frames before and after the frame of interest (i.e., the frame showing the inference), it is likely that the inference (and associated metadata) will be captured and committed to distributed ledger **178** and/or database **180** for authenticity.
(74) Other configurations can include entries with more or less frames per entry, and the number of frames can be dependent upon the inference that is determined/discovered by AI model subscribers **126**A and **126**B. For example, if the inference lasts a duration of time that is shown in forty frames, the entry to the distributed ledger **178** and/or database **180** can be fifty frames (which includes five frames before and five frames after the frames encompassing the inference).
(75) Before being committed to distributed ledger **178** and/or database **180**, the entry can be assigned an identification number. The identification number can be saved in a storage media, such as database **180**, to enable identification and access to the entry as saved in distributed ledger **178**. The identification number should be the same number for the entry if it is committed/saved in multiple locations, and different entries should have different identification numbers than previous entries. Therefore, with each entry having a different identification number (and the number being saved in searchable database **180**), each entry is easily identifiable and locatable in distributed ledger **178**. The identification number can be assigned by/at sink **176** or by another component of system **110**, such as a computer processor.
(76) Distributed ledger **178** can be a blockchain ledger that is public or private. Distributed ledger **178** can be a database with one or multiple storage media that allows for entries (that can include video data **121**A, **121**B, and/or **21** and associated metadata) to be committed/saved and shared across multiple networks, sites, institutions, geographies, processors, etc. (i.e., nodes). The participant at each node of distributed ledger **178** can access the entries committed/saved/shared across distributed ledger **178** and can own an identical copy of the entry. Any additions (i.e., new entries) to distributed ledger **178** are reflected and copied to all nodes. The nodes communicate to ensure that each entry is identically saved at each node. Thus, manipulation of the entries within distributed ledger **178** is difficult as each entry saved at each node would have to be manipulated without detection by distributed ledger **178**.
(77) Sink **176** can include customer **184**, which can include a controller, computer processor, digital storage media, customer interface, and/or other components. Customer **184** can have access to all of the data coming to (and accessible by) sink **176**, such as first annotated AI video data **121**A, second annotated AI video data **121**B, and original video data **21** (and associated metadata). For example, the customer interface of customer **184** can be configured to view first annotated video data **121**A, second annotated video data **121**B, and/or original view data **21** (and associated metadata). Additionally, a node of distributed ledger **178** can be located at and/or under the control of customer **184** such that any information/entries committed to distributed ledger **178** is also accessible by customer **184**. Thus, customer **184** can evaluate and/or analyze video data **121**A, **121**B, and/or **21** and associated metadata to confirm/ensure the inferences are correct. In such a situation, because customer **184** is under the control of a node of distributed ledger **178**, customer **184** would have to authorize the commitment of the entry to distributed ledger **178**. Therefore, customer **184** can play a role in the confirmation of inferences and the commitment of entries to distributed ledger **178**.
(78) Database **180** can include a computer processor, digital storage media, and/or other components that allowed for the organized collection of structured data, such as entries that includes video data **121**A, **121**B, and/or **21** and associated metadata (and potentially identification numbers for each entry). Database **180** can be modeled in rows and columns in a series of tables to make identifying and accessing the entries and the information associated with the entries (video data and/or metadata) easy and organized, though any type of database usable for storing the information in an organized manner for later retrieval can be utilized. Database **180** can be a

separate component, or can be combined with sink **176**, blockchain subscriber **126**C, and/or other components of system **110**. Database **180** is configured to allow access and viewability of each entry without the need to access distributed ledger **178**. Thus, a party can look at the entries in database **180** as a preliminary analysis of the entry and then refer to the entry in distributed ledger **178** for an authenticated version of the entry. Because the entry can have the same identification number in database **180** as in distributed ledger **178**, the entry in distributed ledger **178** is easy to find as it corresponds to the entry in database **180**. Additionally, database **180** can include other information for identifying the entry in distributed ledger **178**, such as the date and time that the entry was committed to distributed ledger **178**. Further, database **180** can include or function in association with applications **182**, which can be configured to sort, analyze, edit, process, and/or otherwise examine entries in database **180** for any desired purpose.
(79) The entries committed to distributed ledger **178** do not need verification from any other system, and can be stand-alone data that does not need information or reference to any other system. Thus, if the entries need to be accessed and analyzed long after being committed to distributed ledger **178**, no further information or context is needed by the analyzing party to fully understand and review video data **121**A, **121**B, and/or **21** (and associated metadata). The entry in distributed edger **178**, even when accessed long after the entry is committed, is credible and authentic due to the characteristics of distributed ledger **178** and the commitment process of the entry (which includes video data **121**A, **121**B, and **21** (and associated metadata)) by system **110**. Thus, the authenticity of first annotated video data **121**A, second annotated video data **121**B, and/or original video data **21** can be confirmed by reviewing the entry of that video data in distributed ledger **178**.
(80) System **110** can include other components and capabilities not expressly disclosed herein. For example, system **110** can have an audit system that reports data to customer **184** continuously or periodically. In one example, audit system can report to customer **184** (or other interested parties) a summary of inferences determined by AI model subscribers **126**A and **126**B and/or entries committed to distributed ledger **178**. In another example, audit system can report a list of the entries committed to distributed ledger **178** along with the identification number for each entry.
(81) While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A method of committing video data and associated metadata to a blockchain ledger, the method comprising: preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model; accessing the preprocessed video data and associated metadata by the first artificial intelligence model, the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data; formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference; accessing the preprocessed video data and associated metadata by a blockchain subscriber independent from the first artificial intelligence model; in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, committing a first entry to the blockchain ledger that

includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data, wherein the blockchain ledge includes multiple nodes each having digital storage media and the entry is committed for storage at each node; committing the first entry to a first node of the multiple nodes of the blockchain ledger, the first node being located at a customer terminal; and accessing at least one of the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data at the customer terminal.

2. The method of claim 1, further comprising: assigning an identification number to the first entry.

3. The method of claim 2, further comprising: accessing the first entry in the blockchain ledger using the identification number.

4. The method of claim 3, further comprising: comparing the first annotated video data and metadata associated with the first annotated video data to the video data and metadata associated with the video data of the first entry in the blockchain ledger to confirm an authenticity of the first annotated video data and metadata associated with the first annotated video data.

5. The method of claim 1, further comprising: in response to the determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, saving the first entry to a database that includes digital storage media.

6. The method of claim 5, further comprising: assigning an identification number to the first entry that is both committed to the blockchain ledger and saved to the database.

7. The method of claim 1, wherein the step of accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the video data comprises: monitoring, by the first artificial intelligence model, the preprocessed video data for the occurrence of the first inference; and formulating the first output in response to an occurrence of the first inference.

8. The method of claim 1, wherein the video data shows at least a portion of a manufacturing system for constructing a product and the first inference occurs in response to a determination by the first artificial intelligence model that the product is incorrectly constructed.

9. The method of claim 1, further comprising: accessing the preprocessed video data and associated metadata by a second artificial intelligence model, the second artificial intelligence model being configured to determine a second output that is indicative of a second inference dependent upon the preprocessed video data; formulating second annotated video data and associated metadata by the second artificial intelligence model including the second output indicative of the second inference; and in response to a determination by the second artificial intelligence model that the second inference has occurred so as to prompt the formulation of the second output, committing a second entry to the blockchain ledger that includes the second annotated video data, metadata associated with the second annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data.

10. A system for authenticating video data and metadata associated with the video data, the system comprising: a blockchain ledger; a gateway, which includes a first computer processor, that is configured to preprocess the video data according to preprocessing parameters defined within a runtime configuration file to edit the video data to be suitable for use by a first artificial intelligence model; the first artificial intelligence model configured to: access the preprocessed video data and associated metadata, determine a first output that is indicative of a first inference dependent upon the preprocessed video data, and formulate first annotated video data and associated metadata that includes the first output indicative of the first inference; a blockchain subscriber, which is executed by a second computer processor, that is independent from the first artificial intelligence model and is configured to access the preprocessed video data and associated metadata; a sink in communication with the first artificial intelligence model and the blockchain subscriber, the sink

being configured to, in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, commit a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data; and a second artificial intelligence model configured to access the video data and associated metadata, determine a second output that is indicative of a second inference dependent upon the video data, and formulate second annotated video data and associated metadata that includes the second output indicative of the second inference, wherein the sink is in communication with the second artificial intelligence model and configured to, in response to a determination by the second artificial intelligence model that the second inference has occurred so as to prompt formulation of the second output, commit a second entry to the blockchain ledger that includes the second annotated video data, metadata associated with the second annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data.
11. The system of claim 10, further comprising: a database having digital storage media, wherein the sink is configured to, in response to the determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, save the first entry to the database.
12. The system of claim 11, further comprising: a third computer processor in communication with the sink, the third computer processor being configured to assign an identification number to the first entry committed to the blockchain ledger and saved to the database.
13. The system of claim 11, wherein the sink includes the third computer processor such that the sink assigns the identification number to the first entry.
14. The system of claim 11, further comprising: a fourth computer processor in communication with the database and configured to access the first entry in the database to view at least one of the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and the metadata associated with the preprocessed video data.
15. The system of claim 10, further comprising: a customer terminal able to access the first annotated video data and metadata associated with the first annotated video data.
16. The system of claim 10, wherein the video data shows at least a portion of a manufacturing system for constructing a product and the first inference occurs in response to a determination by the first artificial intelligence model that the product is incorrectly constructed.
17. The system of claim 10, wherein the first artificial intelligence model and the second artificial intelligence model access and analyze the video data simultaneously to determine the occurrence of the first inference and the second inference, respectively.
18. A method of committing video data and associated metadata to a blockchain ledger, the method comprising: preprocessing the video data, by a first computer processor, according to preprocessing parameters defined within a runtime configuration file, wherein the video data is preprocessed to edit the video data to be suitable for use by a first artificial intelligence model; accessing the preprocessed video data and associated metadata by the first artificial intelligence model, the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the preprocessed video data; formulating first annotated video data and associated metadata by the first artificial intelligence model that includes the first output indicative of the first inference; accessing the preprocessed video data and associated metadata by a blockchain subscriber independent from the first artificial intelligence model; in response to a determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, committing a first entry to the blockchain ledger that includes the first annotated video data, metadata associated with the first annotated video data, the preprocessed video data, and metadata associated with the preprocessed video data; assigning an identification number to the first entry; accessing the first entry in the blockchain ledge using the identification number; and comparing the first annotated video data and metadata associated with

the first annotated video data to the video data and metadata associated with the video data of the first entry in the blockchain ledge to confirm an authenticity of the first annotated video data and metadata associated with the first annotated video data.

19. The method of claim 18, wherein the step of accessing the preprocessed video data and associated metadata by the first artificial intelligence model with the first artificial intelligence model being configured to determine a first output that is indicative of a first inference dependent upon the video data comprises: monitoring, by the first artificial intelligence model, the preprocessed video data for the occurrence of the first inference; and formulating the first output in response to an occurrence of the first inference.

20. The method of claim 18, further comprising: in response to the determination by the first artificial intelligence model that the first inference has occurred so as to prompt the formulation of the first output, saving the first entry to a database that includes digital storage media.